EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 9B

1. Echoes in a Room: Canceling the Echo

Consider a simple acoustic model of a room that produces one echo. The echo arrives 3 samples after the original sound and has amplitude 0.6 relative to the direct sound.

Thus, the impulse response of the room is

$$h[n] = \delta[n] + 0.6\,\delta[n-3].$$

A short audio signal is given by

$$x[n] = \begin{cases} 1, & n = 0, \\ 0, & n = 1, \\ -1, & n = 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Plot the impulse response $h[n]$.

(b) Write an expression for $y[n] = (x * h)[n]$ in terms of $x[n]$. Then compute $y[0], \ldots, y[6]$ for the short audio signal above.

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(c) We now want to remove the echo from a recording $y[n] = x[n] + 0.6x[n-3]$, where $x[n]$ may be any signal that is zero for $n < 0$. Someone proposes subtracting a delayed, scaled copy of the recording:

$$z[n] = y[n] - 0.6y[n-3].$$

Express $z[n]$ in terms of $x[n]$. Does this recover the original signal exactly? Describe the remaining distortion.

(d) Instead, consider a system that uses its own previous output:

$$z[n] = y[n] + b\,z[n-3], \qquad z[n] = 0 \text{ for } n < 0.$$

Choose b so that $z[n] = x[n]$ for every input signal described in part (c). Explain why your choice removes the echo exactly.

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2. Impulse Response of an IIR Filter

A system can feed a fraction of its previous output back into its current output, allowing the effect of a single input to persist. For example, if a microphone is brought too close to its speaker, sound that the microphone picks up will reverberate.

Consider the discrete-time LTI system

$$y[n] = \frac{1}{2}y[n-1] + x[n].$$

Assume the system starts at rest: $x[n] = y[n] = 0$ for $n < 0$.

(a) Let $x[n] = \delta[n]$. Use the recurrence to compute $y[0], \ldots, y[5]$ and sketch these samples. Identify an expression for the impulse response $h[n]$ for all integers n.

(b) Why does the output remain nonzero after the impulse has ended? Now consider replacing the coefficient $\frac{1}{2}$ by $\frac{3}{4}$, and then by $-\frac{1}{2}$. For each replacement, write the first four impulse-response samples and describe how the response differs from part (a). Does the original response ever become exactly zero? Explain why this is called an *infinite impulse response* (IIR) filter.

(c) Return to the original coefficient $\frac{1}{2}$. Let $x[n] = 1$ for $n = 0,\ 1,\ 2$ and zero otherwise. Using the impulse response from part (a), compute the output through the convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

Sketch $x[k]$ and $h[n-k]$ as functions of k for $n = 1$ and $n = 3$. Where do nonzero samples from each signal overlap? Then calculate $y[0], \ldots, y[5]$

Rewrite $y[n]$ as a sum of shifted copies of $h[n]$. Why does the output continue after the input ends?

(d) **When is feedback stable?** A system is *bounded-input, bounded-output (BIBO) stable* if every bounded input produces a bounded output. You may use the following fact: a DT-LTI system is BIBO stable exactly when its impulse response is *absolutely summable*, meaning

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$

Replace the coefficient $\frac{1}{2}$ in the original system by a real constant a:

$$y[n] = a\,y[n-1] + x[n],$$

with the system initially at rest. Find its impulse response $h_a[n]$ and determine exactly which values of a make the system BIBO stable. Justify your answer using the absolute sum.

(e) **A bounded response is not enough.** Set $a = -1$. A unit-step input produces the bounded output $1, 0, 1, 0, \ldots$, even though part (d) says this system is unstable. Find a bounded input, zero for $n < 0$, that produces an unbounded output. Give an expression for that output and explain why this demonstrates BIBO instability.